

HALF ADDER & FULL ADDER

Digital Logic Design using Logisim

Half Adder | Full Adder | Truth Tables | Logisim Simulation

KLH-FED-2026-Section-2--Batch-8

[github.com/Keerthan858/KLH-FED-2026-Section-2--Batch-](https://github.com/Keerthan858/KLH-FED-2026-Section-2--Batch-8)

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1. What is a Binary Adder?

A combinational circuit that adds binary numbers.

SUM is the result bit; CARRY is the overflow.

A core building block of ALUs, CPUs and calculators.

Two common circuits: Half Adder and Full Adder.

2. Half Adder

Adds two 1-bit binary inputs: A and B

Inputs: A, B

Outputs: Sum (S), Carry (C)

SUM is produced by XOR: $S = A \oplus B$

CARRY is produced by AND: $C = A \cdot B$



3. Half Adder Truth Table

A	B	SUM ($A \oplus B$)	CARRY ($A \cdot B$)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

4. Half Adder in Logisim

Build and test the circuit

Place two Input Pins (A, B).

Add an XOR gate (SUM) and an AND gate (CARRY).

Connect A and B to both gates.

Attach Output Pins/LEDs to SUM and CARRY.

Toggle inputs and check all 4 truth-table cases.

5. Full Adder

Adds A, B and Carry-In (Cin)

Inputs: A, B, Cin

Outputs: SUM and Carry-Out (Cout)

$$\text{SUM} = A \oplus B \oplus \text{Cin}$$

$$\text{Cout} = AB + \text{Cin}(A \oplus B)$$

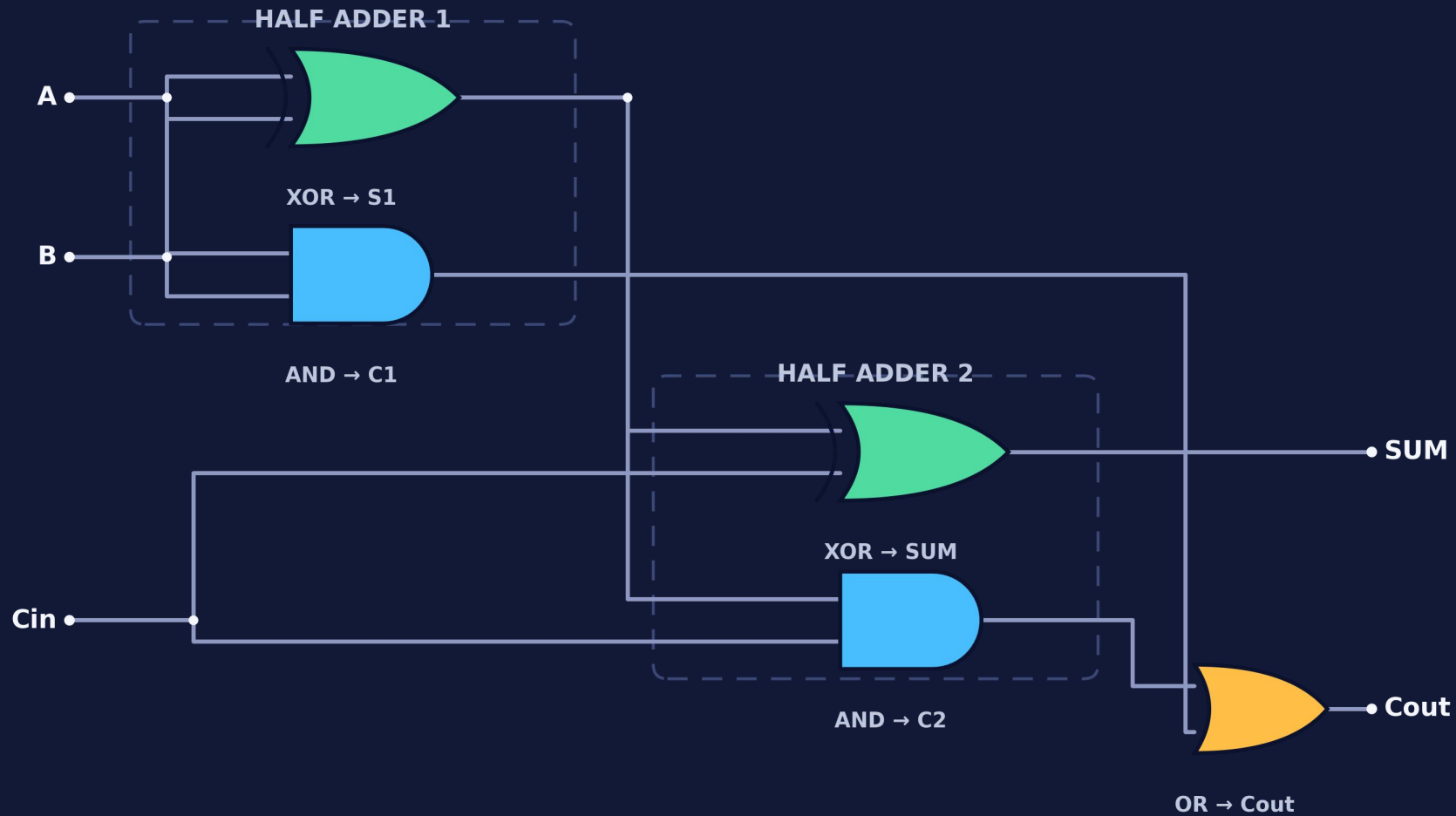
Built from two Half Adders + one OR gate.

6. Full Adder Truth Table

A	B	Cin	SUM	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

7. Full Adder using Two Half Adders

Two Half Adders + one OR gate — modular, easy to build, test and expand



8. Full Adder in Logisim

Recommended component layout

Inputs: A, B, Cin (3 Input Pins).

Create HA1 with XOR + AND.

Feed HA1 SUM into HA2 with Cin.

OR the two carry outputs to get Cout.

Two Output Pins/LEDs for SUM and Cout.

Test all 8 input combinations.

9. Half Adder vs Full Adder

Feature	Half Adder	Full Adder
Inputs	2 (A, B)	3 (A, B, Cin)
Outputs	SUM, Carry	SUM, Cout
Carry-In	Not supported	Supported
Typical use	LSB/basic addition	Multi-bit binary addition
Expansion	Limited	Can be cascaded

10. Key Takeaways

Half Adder = XOR + AND

Full Adder = two Half Adders + OR

Carry-In lets Full Adders chain for multi-bit addition.

Logisim makes gate-level design and testing easy.

Always test every input combination before trusting the circuit.



THANK YOU

Questions?

Half Adder • Full Adder • Logisim